

Arithmetic and complexity

Explicit tensors with quadratic border rank

Difficulty: extreme **Importance:** broadly interesting

Topic: explicit lower bounds for tensor decompositions

Status: open; admitted after a bounded literature search found no resolution.

Context and notation

Over $\mathbb{C}$, the tensor rank $R(T)$ is the minimum number of pure tensors in an exact sum for T . The border rank $\underline{R}_{\mathbb{C}}(T)$ is the least r such that T is a Euclidean limit of complex tensors of rank at most r .

Problem statement

Construct a deterministic algorithm that, on input the positive integer n , outputs the rational entries of a tensor $T_n \in \mathbb{Q}^{n \times n \times n}$ in time polynomial in n , and prove that there are constants $c > 0, n_0$ such that $\underline{R}_{\mathbb{C}}(T_n) \geq cn^2$ for all $n \geq n_0$. Output rationals are encoded by binary integer numerators and denominators; thus the time bound also limits their bit lengths. Border rank is over $\mathbb{C}$, as defined in the notation.

Why it matters

Explicit hard tensors would supply lower bounds for bilinear computation that are much stronger than the currently available examples. Generic existence does not provide the required efficient construction.

References and status

M. Michałek and J. Landsberg, *Towards Finding Hay in a Haystack: Explicit Tensors of Border Rank Greater Than $2.02m$* , Theory of Computing 21(13), 2025, §1 and Definition 1.1, distinguishes polynomial-time explicitness from weaker notions and obtains a linear bound. The quadratic target and rational-output encoding are an editorial quantitative formulation of the paper's explicit-versus-generic challenge, not a numbered conjecture quoted from it. Searches for "explicit tensors superlinear rank 2026" and "explicit tensors superlinear border rank" located no quadratic construction; weaker semi-explicit constructions do not meet this statement. **Admitted: no resolution located.**